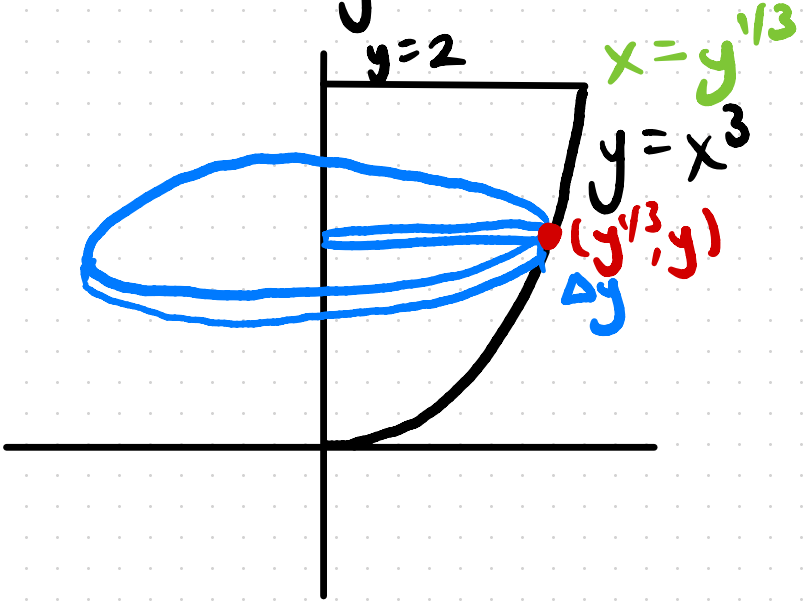
Volumes by cylindrical shells (6.3)

The region bounded by $x=0$, $y=2$, $y=x^3$ is rotated about the y-axis.



Integrate volumes of disks in y-direction

Volume of disk is

$\pi (\text{radius})^2 \Delta y$

↑

want radius in terms of y, since we will integrate in y-direction.

radius: $y^{1/3}$

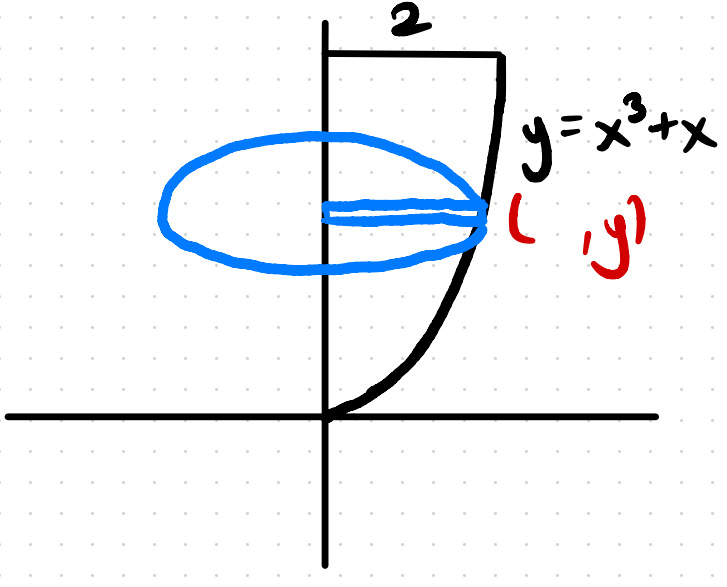
$$\int_0^2 \pi (y^{1/3})^2 \, dy = \int_0^2 \pi y^{2/3} \, dy$$

$$= \pi \cdot \frac{3}{5} y^{5/3} \Big|_0^2$$

$$= \pi \cdot \frac{3}{5} 2^{5/3}$$

$$= \frac{6}{5} \pi 2^{2/3}$$

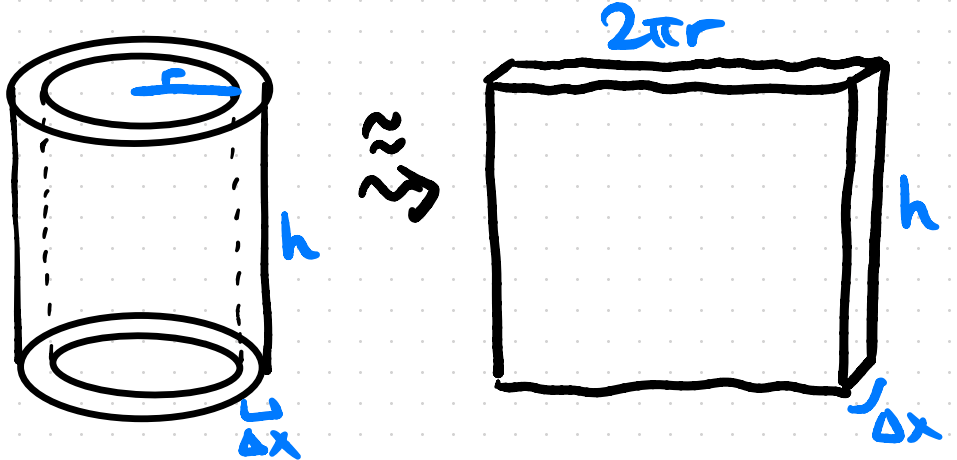
The region bounded by $x=0$, $y=2$, $y=x^3+x$ is rotated about the y-axis.



radius of cross-section = ???
in terms of y

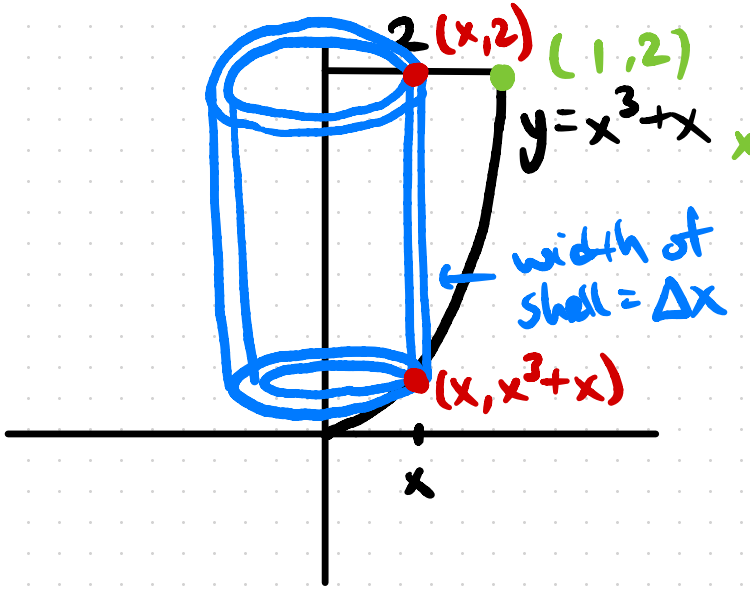
The method of cylindrical shells is a technique used specifically for finding volumes of solids formed by rotating around an axis.

We decompose our solid into hollow cylindrical "shells" with width Δx, where x is the direction perpendicular to the axis of rotation.



Volume of shell $\approx$

$$2\pi r h \Delta x$$



At a particular x-value,

radius of shell $= x$

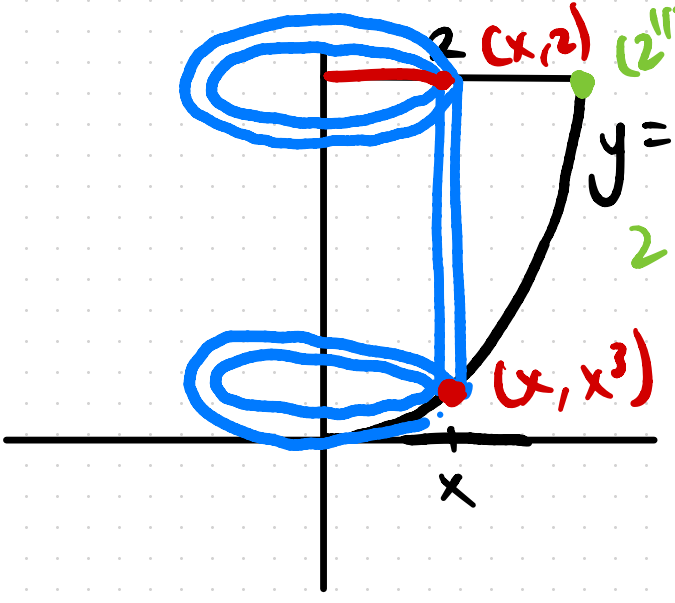
height of shell $= 2 - x^3 - x$

Volume of shell $\approx 2\pi x(2-x^3-x)\Delta x$

Volume of solid $= \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi x_i^*(2-x_i^{*3}-x_i^*)\Delta x$

where we subdivide into n intervals and choose x_i^* from each interval

$$= \int_0^1 2\pi x(2 - x^3 - x)\,dx$$



radius $= x$

height $= 2-x^3$

$$\int_0^{2^{1/3}} 2\pi x(2-x^3)\,dx$$

$$= 2\pi\int_0^{2^{1/3}} (2x-x^4)\,dx$$

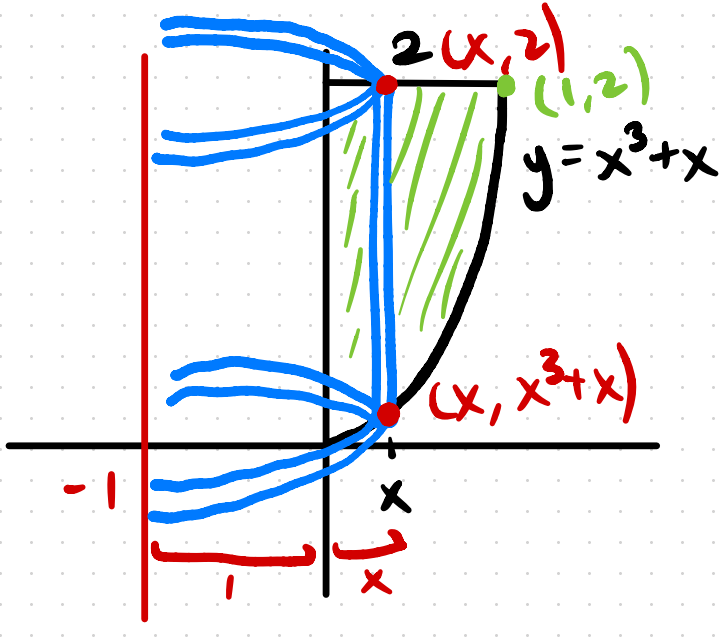
$$= 2\pi\left(x^2-\frac{1}{5}x^5\right)\Big|_0^{2^{1/3}}$$

$$= 2\pi\left(2^{2/3} - \frac{1}{5}2^{5/3}\right)$$

$$= 2\pi\left(2^{2/3} - \frac{2}{5}2^{2/3}\right)$$

$$= \frac{6}{5}\pi 2^{2/3}$$

The region bounded by $x=0$, $y=2$, $y=x^3+x$ is rotated about the line $x=-1$.



radius $= 1+x$

height $= 2-x^3-x$

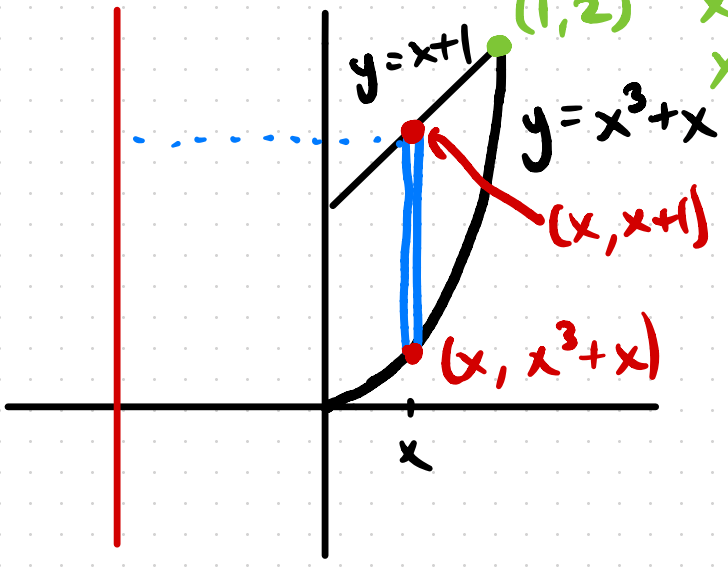
$$\int_0^1 2\pi(1+x)(2-x^3-x)\,dx$$

The region bounded by $x=0$, $y=x+1$, $y=x^3+x$ is rotated about the line $x=-1$

$x^3+x=x+1$

$x^3=1$

$x=1$



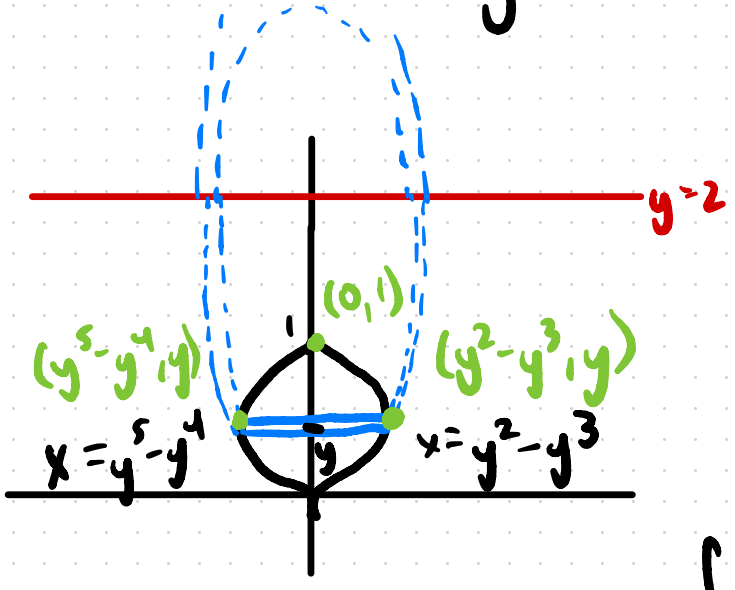
radius $= 1+x$

height: $(x+1)-(x^3+x)$

$= 1-x^3$

$$\int_0^1 2\pi(1+x)(1-x^3)\,dx$$

The region bounded by $x = y^5 - y^4$, $x = y^2 - y^3$ is rotated about the line $y = 2$.

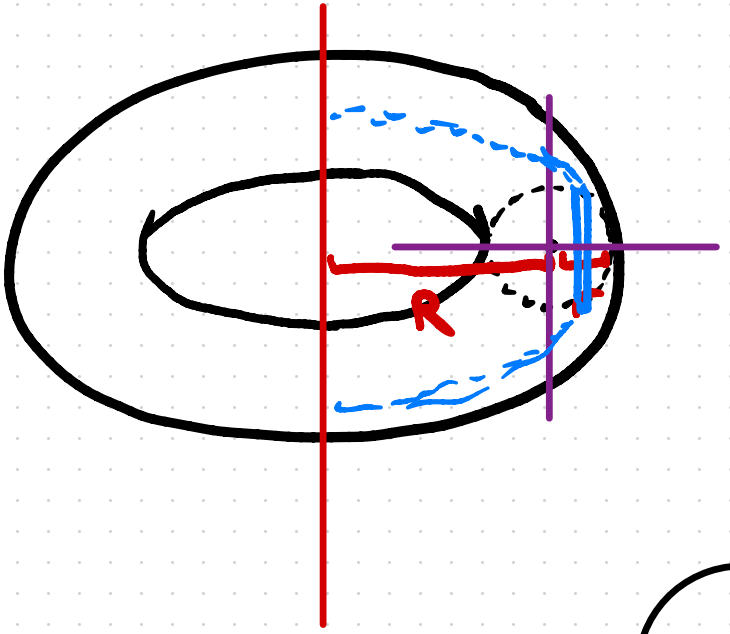


$$\text{radius} = 2 - y$$

$$\text{height} = (y^2 - y^3) - (y^5 - y^4)$$

$$= y^2 - y^3 + y^4 - y^5$$

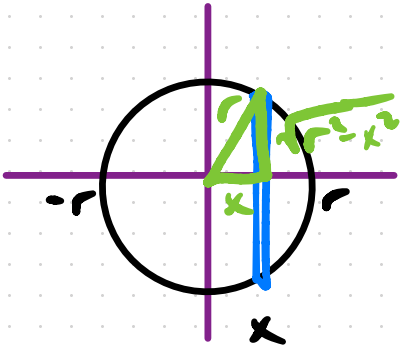
$$\int_0^1 2\pi(2-y)(y^2 - y^3 + y^4 - y^5)\,dy$$



radius of shell $= R + x$

height $= 2\sqrt{r^2 - x^2}$

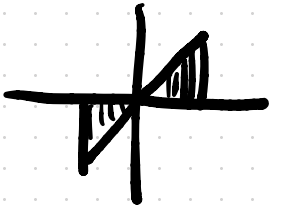
solid torus



$x^2 + y^2 = r^2$

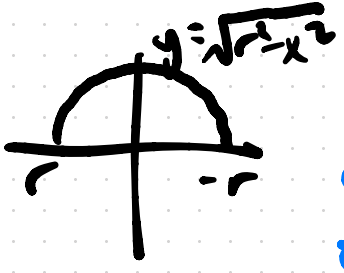
$y = \pm\sqrt{r^2 - x^2}$

$$\int_{-r}^{r} 2\pi(R+x)\,2\cdot\sqrt{r^2-x^2}\,dx$$



$$= 4\pi R\int_{-r}^{r}\sqrt{r^2-x^2}\,dx + 4\pi R\int_{-r}^{r} x\sqrt{r^2-x^2}\,dx$$

$$= 0$$



area of a semicircle of radius r

$$\frac{1}{2}\pi r^2$$

$$= 4\pi R \cdot \frac{1}{2}\pi r^2$$

$$= 2\pi^2 R r^2$$